

Penetration Testing Tales

Estimated time needed: 4 minutes

Objectives

After completing this lab, you will be able to:

- Describe real-world scenarios in which pen testing was carried out in different ways, including using hidden cameras, breaking and entering, jumping the fence, and assessing lunch breaks

Introduction

A penetration test is a security test that launches a mock cyberattack to find vulnerabilities in a computer system. It is conducted by skilled security professionals who use hacking tools and techniques to identify security weaknesses in an organization's computing systems.

Penetration testing, also known as pen testing, can be carried out in different ways. Let's explore some real-life scenarios that involve pen testing.

Hidden cameras

During a pen test for a major manufacturing company, the tester discovered a network of hidden security cameras set up throughout the facility. These cameras were internet-accessible for remote viewing purposes. The pen testing team test identified a zero-day vulnerability in the camera software. Using this vulnerability, they bypassed authentication and gained access to approximately 20 cameras. Many of these cameras were pointed at workstations, providing the testers with a unique opportunity to zoom in and capture valid credentials as employees logged in. The team successfully used these credentials to gain external access to the company's network.

Breaking and entering

Coalfire penetration testers Gary De Mercurio and Justin Wynn were hired by Iowa's Judicial Branch to test the security of the Dallas County Courthouse. As a part of their physical penetration test, the two men broke into the courthouse when they unknowingly tripped an alarm. When the police arrived, the men told the authorities that they had been hired to test the security. Despite verification from the state official who had hired them, they were arrested and faced felony charges. Over the next month, questions arose over the jurisdiction and the scope of their contract. Ultimately, the two men were exonerated, and Iowa implemented new policies regarding information security contracts.

Jumping the fence

An energy company hired a pen tester. After performing reconnaissance on the building, the pen tester determined that an 8-foot barbed-wire fence surrounded the location. One evening, taking advantage of the darkness, he jumped the fence and explored the property, avoiding the cameras. He was surprised to find an unlocked commercial vehicle with a laptop inside. Excited by the possibilities, he requested permission from his contact at the organization and took the laptop. He bypassed the login, added a local administrator user, and planted malware on the laptop to maintain control. He accessed the corporate network and gained local admin access on other systems, ultimately gaining domain admin access. He then had complete control over the company's network and systems.

Lunch break

An electrical distribution plant hired a pen tester. On the first day of the test, the tester spent most of the day at the main office. He watched to see how employees dressed, what time they showed up to work, when they took lunch, how long their lunch was, and whether they were usually alone or in groups. The next day, he acquired a large shipping box and posed as a deliveryman. He waited for the lunch hour when many people entered and exited the building. He held the box in one hand and his phone in the other. When he got to the door, he asked someone to keep it open as they walked out. Without a problem, he was able to enter the building. He took pictures, found sensitive files, grabbed a few laptops, and exited the building. His next goal was to access the substation surrounded by a 12-foot tall chain link fence. While the top of the fence was secured with razor wire, the bottom of the fence was just sitting on gravel. He found a spot where the gravel had washed out a bit and, with a bit of effort, could crawl under the fence where he could access the control shed and the security camera recordings.

Summary

In this reading, you learned that:

- A penetration test is a security test that launches a mock cyberattack to find vulnerabilities in a computer system.
- It has been carried out in different ways, including using hidden cameras, breaking and entering, jumping the fence, and assessing lunch breaks with predefined criteria.

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